

get_wifi

REFERENCE

API consumer

read the reader's saved Wi-Fi profiles and connection status

Overview

`get_wifi` retrieves the **Wi-Fi profiles saved on the reader** together with their current connection status and configuration. The response is a list (`wifiProfiles.wifiConfig[]`) of profiles; each describes the wireless interface, the associated access point (ESSID, security type, connection status), the IPv4/IPv6 addressing, and a `securityDetails` block keyed by the security type (`WPA2Personal` , `WPA3Personal` , `WPA2Enterprise` , `WPA3Enterprise`). It is a read-only query on the device-management (MGMT) endpoint.

The device **masks** stored credentials — passwords, passphrases, and identities are returned as a mask, never the real secret. Use `get_wifi` to verify which SSID a reader is connected to and confirm its addressing before relying on the wireless uplink.

Details

Property	Value
Interaction Pattern	Request / Response (app publishes a command; device publishes the correlated reply)
State Change	READ-ONLY
Required Request Fields	<code>command</code> , <code>requestId</code>
Applies To	Endpoint: Management (MGMT). Reader: RFD40 series, RFD90 series
API Version	<code>v1.1</code> (API Version Reporting)
Related	set_wifi , delete_wifi_profile , get_eth

When to Use

Use `get_wifi` when you need to **verify the reader's wireless configuration** — during commissioning, a connectivity check before inventory, or a fleet audit of which SSID each device is on.

- **Confirm the connection:** check `accessPoint.status` (CONNECTED?), `accessPoint.essid` (correct network?), and `accessPoint.isPreferred`.

- **Validate security:** check `accessPoint.securityType` matches the AP; the matching `securityDetails` block shows the (masked) credential/protocol configuration.
- **Validate addressing:** check `ipv4Configuration.ipAddress`, `gateway`, and `enableDhcp` to confirm the reader has a working route to the broker / back-end.

Topic Mapping

Message	Direction	Topic	Note
Request	App → Reader	<code><tenantId>/MGMT/clients/cmnd/<serial></code>	On the management endpoint.
Response	Reader → App	<code><tenantId>/MGMT/clients/resp/<serial></code>	Correlated reply to the request above; the in-payload <code>requestId</code> links the twin topics.

Topics follow the pattern `<tenantId>/<topic>/<serial>` — the tenant id assigned to you at provisioning (a slot you fill in, never a literal), the base topic you configure on the endpoint, and the reader's serial number, which the reader appends itself in its model-prefixed form (for example `RFD40-212735201D0053`). `App → Reader` means your app publishes and the reader subscribes; `Reader → App` means the reader publishes and your app subscribes.

Credentials are presented once at CONNECT. The `requestId` you send is echoed in the response so you can pair reply to request. The endpoint's `mqttParams` govern QoS and retain, not this command.

See [Topics and Channels](#).

Key Fields

- `requestId` (request) — your correlation token; make it unique per request. The device echoes it in the response so you can pair reply to request.
- `wifiProfiles.wifiConfig[]` (response) — the array of saved profiles; iterate it to find the connected one.
- `...accessPoint.status` / `essid` (response) — the association state and network name of each profile.
- `...accessPoint.securityType` (response) — selects which `securityDetails` key is populated.
- `...ipv4Configuration.ipAddress` (response) — the DHCP-assigned address the reader uses to reach the broker.

- `response.code` (response) — read this to determine success vs failure (see [Response Codes](#)).

Request Payload

The request envelope is just `command` and `requestId` — there is no payload wrapper. Use the **Schema** tab for the field reference and the **Example** tab for a ready-to-send, copy-pasteable request.

Request payload

```
1 {
2   "command" : "get_wifi" ,
3   "requestId" : "abc123"
4 }
```

Request payload

* required • — not applicable • Colored badge = JSON type

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
1	command * and	string	—	get_wifi	Fixed command name. Retrieves the Wi-Fi profiles saved on the device.
2	requestId * d	string	—	abc123	A unique identifier for the request, allowing tracking and debugging. Echoed back unchanged in the response.

Response Payload

The response envelope is `command`, `requestId`, `apiVersion`, `response`, and a `wifiProfiles` object whose `wifiConfig` is the **array of profiles**. On the **Schema** tab, expand `wifiProfiles` → `wifiConfig` → `interfaceDetails` (and its `accessPoint`, `ipv4Configuration`, `ipv6Configuration`, `securityDetails` sub-objects) to see every field. Within each profile's `interfaceDetails`, only `interfaceName`, `hostname`, and `macAddress` are required; the interface-level `status` is optional. `securityDetails` carries only the key matching `accessPoint.securityType`, with secrets masked. The **Example** tab shows a WPA2-Personal profile; enterprise variants are in [Examples](#).

Connection scenario

WPA2 Personal connection ▼

Response payload

Example: WPA2-Personal

```
1  {
2    "command" : "get_wifi" ,
3    "requestId" : "abc123" ,
4    "apiVersion" : "V1.1" ,
5    "wifiProfiles" : {
6      "wifiConfig" : [
7        {
8          "interfaceDetails" : {
9            "interfaceName" : "wlan0" ,
10           "status" : "" ,
11           "hostname" : "RFD40-211545201D0034" ,
12           "macAddress" : "AC:3F:A4:00:2B:15" ,
13           "accessPoint" : {
14             "essid" : "IOT_TESTAP" ,
15             "status" : "CONNECTED" ,
16             "securityType" : "WPA2Personal" ,
17             "isPreferred" : false ,
18             "autoConn" : false
19           } ,
20           "ipv4Configuration" : {
21             "ipAddress" : "192.168.0.110" ,
22             "subnetMask" : "255.255.255.0" ,
23             "gateway" : "192.168.0.1" ,
24             "enableDhcp" : true ,
25             "dnsServer" : "192.168.0.1" ,
26             "domainName" : "zebra.com"
27           } ,
28           "securityDetails" : {
29             "WPA2Personal" : {
30               "password" : "XXXXXXXX"
31             }
32           }
33         }
34       ]
35     } ,
36     "response" : {
37       "code" : 0 ,
38       "description" : "Success"
39     }
40   }
41 }
```

Response payload

* required • — not applicable • Colored badge = JSON type

#	FIELD	TYPE	ENUM / UNITS	EXAMP
1	command *	string	—	get_v
2	requestId *	string	—	abc12
3	apiVersion *	enum	V1.0 V1.1 V1.2 V1.21 V1.22	V1.1
4	wifiProfiles	object	—	—
5	wifiConfig	array	—	—
6	interfaceDe tails	object	—	—
7	interfa ceName *	string	—	wlan0
8	status	enum	Enabled Disabled	Enabl
9	hostnam e *	string	—	RFD40
10	macAddr ess *	string	—	E0-D0

#	FIELD	TYPE	ENUM / UNITS	EXAMP
11	accessPoint	object	—	—
12	essid *	string	—	Zwire
13	status *	enum	CONNECTED DISCONNECTED	CONN
14	securityType *	enum	WPA2Personal WPA3Personal WPA2Enterprise WPA3Enterprise	WPA2I
15	isPreferred *	boolean	—	false
16	autoConn	boolean	—	false
17	ipv4Configuration	object	—	—
18	ipAddress	string	ipv4	192.1
19	subnetMask	string	ipv4	255.2
20	gateway	string	ipv4	192.1
21	enableDhcp	boolean	—	true
22	dnsServer	string	ipv4	192.1
23	domainName	string	—	test.
24	ipv6Configuration	object	—	—
25	ipAddress	string	ipv6	fe80:

#	FIELD	TYPE	ENUM / UNITS	EXAMP
26	prefix	integer	—	64
27	gateway	string	ipv6	2a02
28	enableAuto	boolean	—	true
29	dnsServer	string	ipv6	2001:
30	domainName	string	—	test.
31	securityDetails	object	—	—
32	WPA2Personal	object	—	—
33	password	string	password	test0
34	WPA3Personal	object	—	—
35	password	string	password	test0
36	WPA2Enterprise	object	—	—
37	authentication	enum	tls ttls peap	tls

#	FIELD	TYPE	ENUM / UNITS	EXAMP
38	innerAuthentication	enum	none mschapv2	mscha
39	identity	string	—	test
40	anonymousIdentity	string	—	test
41	password	string	password	test
42	phrase	string	—	test
43	certificate	array	—	—
44	key	enum	ca_cert client_key client_cert cert_key_password	ca_ce
45	value	string	—	wifi_

#	FIELD	TYPE	ENUM / UNITS	EXAMP
46	protocol	enum	WPA2_Enterprise_CCMP	WPA2_
47	WPA3Enterprise	object	—	—
48	authentication	enum	tls ttls peap	tls
49	innerAuthentication	enum	none mschapv2	mscha
50	identity	string	—	test
51	anonymousIdentity	string	—	test
52	password	string	password	test
53	phrase	string	—	test

#	FIELD	TYPE	ENUM / UNITS	EXAMP
54	cert t i f i c a t e	array	—	—
55	k e y	enum	ca_cert client_key client_cert cert_key_password	ca_ce
56	v a l u e	string	—	wpa3_
57	pro to c o l	enum	WPA3_Enterprise_CCMP WPA3_Enterprise_CCMP_256 WPA3_Enterprise_GCMP_128 WPA3_Enterprise_GCMP_256_SHA256 WPA3_Enterprise_GCMP_256_SUITEB_192	WPA3_
58	response *	object	—	—
59	code *	integer	0–101	0
60	description *	string	—	Succ

Response Semantics

- **Credentials are masked.** The device returns stored passwords, passphrases, and identities as a mask (never the real secret). Treat every credential field in the response as opaque — you cannot read back the actual value with `get_wifi`.
- **Security type selects the credential block.** `securityDetails` contains only the one key that matches `accessPoint.securityType` — `WPA2Personal` / `WPA3Personal` (a single `password`) or `WPA2Enterprise` / `WPA3Enterprise` (EAP `authentication`, optional `innerAuthentication` / `identity` / `anonymousIdentity`, a `certificate` array, and a `protocol`).
- `ipv6Configuration` is schema-defined but not returned by the current firmware.

- **Wi-Fi-disabled response shape.** When Wi-Fi is disabled, the contract's own example returns the profile under a **top-level** `wifiConfig` object (rather than `wifiProfiles.wifiConfig[]`), carrying only `interfaceName`, `status` (`Disabled`), `hostname` and `macAddress`. The response schema does not declare a top-level `wifiConfig`, so parse both shapes; the connected shape documented here is `wifiProfiles.wifiConfig[]`.

Response Codes

`get_wifi` returns `response.code`. The codes applicable to this command are below; for the full table see the [Response Codes](#) and [Correlation and Request/Response Pairing](#).

Code	Description
0	Success — Wi-Fi profiles retrieved and returned.
3	Not able to retrieve information (<code>IOT_ERROR_INFO_NOT_AVAILABLE</code>) — the device could not gather the Wi-Fi configuration at this time. Retry after a short delay; if it persists, reboot the device.
101	Default error — generic catch-all.

For the full code list see the [Response Codes](#). (Code `20`, `Wifi is not supported`, applies to the Wi-Fi **setters** `set_wifi` / `set_config`, not to `get_wifi`.)

Examples

`get_wifi` is a single request/response exchange. Publish the standard `{command, requestId}` request shown under [Request Payload](#); the device replies on the response topic with the correlated reply — the `requestId` you send (`abc123`) is echoed back so you can pair reply to request. The **Connection scenario** dropdown under [Response Payload](#) shows each contract example:

- **WPA2 Personal connection** — a single masked `password`.
- **WPA2 Enterprise connection** — an EAP `authentication` method, a `certificate` array of `{key, value}` references and a `protocol`, with credentials masked.
- **WPA3 Enterprise connection** — additionally `innerAuthentication`, `identity` and `anonymousIdentity`, and a WPA3 `protocol` suite.
- **WiFi interface disabled** — the minimal reply for a disabled interface.

Notes and Caveats

`get_wifi` is a read-only getter — safe to poll during commissioning or troubleshooting. Configure profiles with `set_wifi` / `delete_wifi_profile`; for the concept-level view, see [Configure Wi-Fi](#). Enterprise profiles depend on installed certificates — see [The Security Model](#).

Related:  [Configure Wi-Fi](#) ·  `get_eth` ·  [Network and Connectivity](#) ·  [Response Codes](#)